Lecture 2: Propagating uncertainty and shared physics

COMP 521 • Probability, Optimization, and Scientific Discovery • Sections 1.6–1.7

This handout is one part of the full COMP 521 notes. Section, equation, figure and listing numbers match the full notes, and references to anything outside this handout point there. The notation table is in "How to use these notes" (Lecture 1).

1.6 Propagating uncertainty through a model

If $\boldsymbol{\Theta} \sim p(\boldsymbol{\theta})$ and $\boldsymbol{Y} = \mathcal{G}(\boldsymbol{\Theta})$, the distribution of $\boldsymbol{Y}$ is the *pushforward* of $p(\boldsymbol{\theta})$ through $\mathcal{G}$. You rarely need its density; you need to sample it or compute its moments. Three strategies follow.

1.6.1 Worked example: uncertain mass, uncertain luminosity

Take the TOY relation $L/L_\odot = (M/M_\odot)^4$ and $X = M/M_\odot \sim \mathcal{N}(1, 0.05^2)$. (A Gaussian allows negative masses; here that is a 20σ event and harmless, but a general model would truncate.)

First-order (delta method). Linearize: $L \approx f(\mu) + f'(\mu)(X - \mu)$. Then $\mathbb{E}[L] \approx f(\mu) = 1$ and $\sigma_L \approx |f'(\mu)|\sigma = 4 \times 0.05 = 0.20$; equivalently $\sigma_L/L \approx |\mathrm{d}\log L/\mathrm{d}\log M|\,\sigma_M/M = 20\%$.

Exact moments. For a Gaussian, $\mathbb{E}[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4 = 1.01501875$, and the exact standard deviation (from $\mathbb{E}[X^8]$) is 0.2026. So $\mathbb{E}[f(X)] \neq f(\mathbb{E}[X])$: averaging the prediction is not predicting at the average. The shift is set by curvature.

Second-order. Expanding one term further,

$$\mathbb{E}[f(X)] \approx f(\mu) + \tfrac{1}{2} f''(\mu)\,\sigma^2, \tag{11}$$

gives $1 + \frac{1}{2} \cdot 12 \cdot 0.0025 = 1.015$, recovering all but 1.9×10^{-5} of the exact mean (the remaining $3\sigma^4$ is fourth order). In several dimensions the correction is $\frac{1}{2}\,\mathrm{tr}(\mathbf{H}_{\mathcal{G}}\boldsymbol{\Sigma}_\Theta)$ per output.

Monte Carlo. Draw $\boldsymbol{\theta}^{(i)} \sim p(\boldsymbol{\theta})$, evaluate $\mathbf{y}^{(i)} = \mathcal{G}(\boldsymbol{\theta}^{(i)})$, and average: $\widehat{\mathbb{E}[\boldsymbol{Y}]} = N^{-1}\sum_i \mathbf{y}^{(i)}$, with standard error $\sigma_Y/\sqrt{N}$. Monte Carlo needs no derivatives and handles skewed, multimodal and discontinuous outputs, but its error falls only as $N^{-1/2}$, and rare regimes (for example the few stars that reach core collapse) need importance sampling or stratification to be resolved at all.

ex01_propagation.py -- the three strategies side by side, plus a pathwise gradient

```python
"""Uncertainty propagation through the toy relation L/Lsun = (M/Msun)^4.

Compares (i) Monte Carlo, (ii) first-order delta method, (iii) second-order
delta method, and the pathwise gradient of E[L] with respect to the mean mass.
"""
import jax
import jax.numpy as jnp

jax.config.update("jax_enable_x64", True)  # derivative checks need float64


def lum(m):  # toy forward model, valid only in a limited mass range
    return m**4


mu, sigma = 1.0, 0.05
```

```

# (i) Monte Carlo with a FIXED set of standard-normal draws (common random numbers)
eps = jax.random.normal(jax.random.key(0), (1_000_000,))


def mc_mean_lum(mu, sigma):
    return jnp.mean(lum(mu + sigma * eps))  # reparameterization: M = mu + sigma*eps


L_mc = lum(mu + sigma * eps)
print(f"MC     : E[L] = {L_mc.mean():.5f}  sd[L] = {L_mc.std():.5f}"
      f"  (MC std. error of mean = {L_mc.std() / jnp.sqrt(eps.size):.1e})")

# (ii) first-order delta method: mean = f(mu), sd = |f'(mu)| sigma
d1 = jax.grad(lum)(mu)
d2 = jax.grad(jax.grad(lum))(mu)
print(f"Delta-1: E[L] = {lum(mu):.5f}  sd[L] = {abs(d1) * sigma:.5f}")

# (iii) second-order correction to the mean: + (1/2) f''(mu) sigma^2
print(f"Delta-2: E[L] = {lum(mu) + 0.5 * d2 * sigma**2:.5f}")

# Exact Gaussian moments for comparison
exact_mean = mu**4 + 6 * mu**2 * sigma**2 + 3 * sigma**4
print(f"Exact  : E[L] = {exact_mean:.8f}")
z = (L_mc.mean() - exact_mean) / (L_mc.std() / jnp.sqrt(eps.size))
print(f"MC error in units of its standard error: z = {z:+.2f}")

# Pathwise gradient of the sample-average objective vs the exact derivative
g_path = jax.grad(mc_mean_lum, argnums=0)(mu, sigma)
g_exact = 4 * mu**3 + 12 * mu * sigma**2
print(f"dE[L]/dmu: pathwise = {g_path:.5f}   exact = {g_exact:.5f}")
```

Output (executed for these notes):

```
MC     : E[L] = 1.01500  sd[L] = 0.20258  (MC std. error of mean = 2.0e-04)
Delta-1: E[L] = 1.00000  sd[L] = 0.20000
Delta-2: E[L] = 1.01500
Exact  : E[L] = 1.01501875
MC error in units of its standard error: z = -0.11
dE[L]/dmu: pathwise = 4.02994   exact = 4.03000
```

Two habits are worth copying from this listing. First, the Monte Carlo result is reported with its standard error and compared in units of it ($z = -0.11$); a Monte Carlo number without an error bar is not a result. Second, the gradient of the Monte Carlo average with respect to μ is taken with the standard-normal draws held fixed; Section 5.5 explains why that is legitimate here and when it is not.

1.6.2 Linear propagation in many dimensions

Linearizing about the mean $\boldsymbol{\mu}$, $\mathcal{G}(\boldsymbol{\Theta}) \approx \mathcal{G}(\boldsymbol{\mu}) + \mathbf{J}(\boldsymbol{\Theta} - \boldsymbol{\mu})$ with $\mathbf{J} = \partial \mathbf{y}/\partial \boldsymbol{\theta}|_{\boldsymbol{\mu}}$, gives

$$\boldsymbol{\Sigma}_Y \approx \mathbf{J}\, \boldsymbol{\Sigma}_\Theta\, \mathbf{J}^\mathsf{T}. \tag{12}$$

With Stellax and JAX, $\mathbf{J}$ costs a handful of forward- or reverse-mode passes (Section 5.1) instead of thousands of tracks, which is one of the main practical reasons to build a differentiable stellar code. The price is that Eq. (12) assumes $\mathcal{G}$ is close to linear over the width of $p(\boldsymbol{\theta})$.

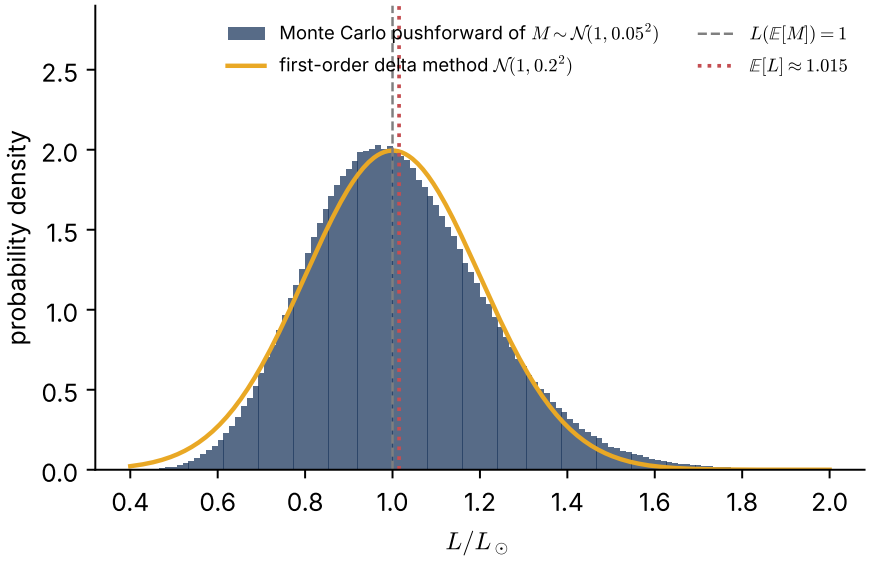


Figure 1: A deterministic nonlinear map turns a symmetric input distribution into a skewed output. The first-order delta method (gold) gets the width nearly right but misses the skew and the shift of the mean. Data are synthetic.

Where linearization breaks

The delta method fails when the input distribution straddles a qualitative change: helium-flash versus quiescent helium ignition, the mass that separates white-dwarf from neutron-star progenitors, a change of convective-core boundary. A Jacobian evaluated on one side knows nothing about the other. Always validate Eq. (12) against a targeted ensemble near such boundaries.

1.6.3 Other propagation methods

Quasi-Monte Carlo replaces random draws with low-discrepancy point sets (for example Sobol sequences); for smooth integrands of modest effective dimension its error falls closer to N^{-1}. *Polynomial chaos expansion* fits the model with polynomials orthogonal under $p(\boldsymbol{\theta})$ and reads moments off the coefficients. Neither is efficient across sharp discontinuities or in very high effective dimension.

Check your understanding

Without computing anything: for $L = M^4$, is $\mathbb{E}[L]$ above or below $L(\mathbb{E}[M])$, and why? Would the answer change for $L \propto M^{0.5}$?

Answer. Above: M^4 is convex, so by Jensen's inequality $\mathbb{E}[f(X)] \geq f(\mathbb{E}X)$. For the concave $M^{0.5}$ the inequality reverses.

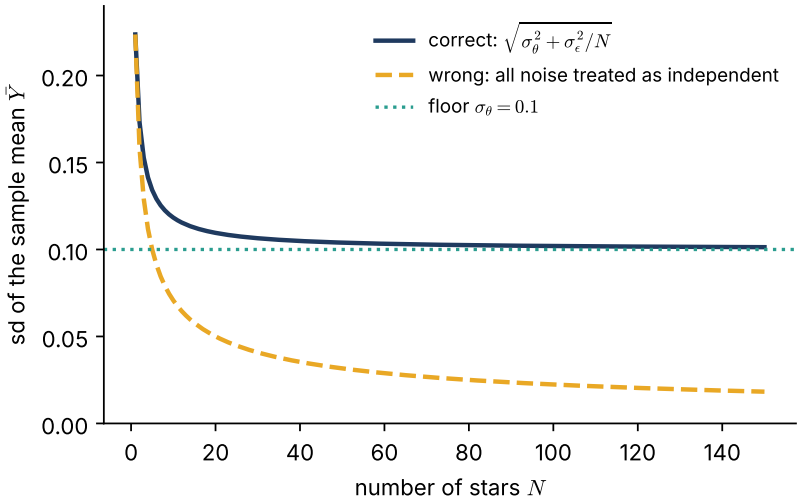


Figure 2: Independent scatter averages down as $N^{-1/2}$; an unconstrained shared offset does not.

1.7 The law of total variance and shared physics

For any random Y and $\boldsymbol{\Theta}$,

$$\mathrm{Var}(Y) = \underbrace{\mathbb{E}_\Theta\big[\mathrm{Var}(Y \mid \boldsymbol{\Theta})\big]}_{\text{scatter at fixed physics}} + \underbrace{\mathrm{Var}_\Theta\big(\mathbb{E}[Y \mid \boldsymbol{\Theta}]\big)}_{\text{spread caused by uncertain physics}} . \tag{13}$$

This splits uncertainty into the part that remains even if the physics were known and the part caused by not knowing it.

Worked example 1.4 (A shared systematic sets a floor)**.** Let $Y_i = \theta + \epsilon_i$ for $i = 1, \ldots, N$, with an uncertain offset shared by all stars, $\mathrm{Var}(\theta) = \sigma_\theta^2$, and independent scatter $\mathrm{Var}(\epsilon_i) = \sigma_\epsilon^2$. For the sample mean $\bar{Y}$, conditioning on θ gives $\mathrm{Var}(\bar{Y} \mid \theta) = \sigma_\epsilon^2/N$ and $\mathbb{E}[\bar{Y} \mid \theta] = \theta$, so Eq. (13) yields

$$\mathrm{Var}(\bar{Y}) = \frac{\sigma_\epsilon^2}{N} + \sigma_\theta^2.$$

With $\sigma_\theta = 0.1$, $\sigma_\epsilon = 0.2$ and $N = 100$, $\mathrm{sd}(\bar{Y}) = \sqrt{0.0004 + 0.01} \approx 0.102$, not the $\sqrt{0.05/100} \approx 0.022$ you would get by treating all noise as independent. Be careful which variance you mean: this is the variance of the *sample mean* $\bar{Y}$. A single star's Y_i has variance $\sigma_\theta^2 + \sigma_\epsilon^2$, which does not depend on N at all.

More stars cannot average away a shared parameter, but more *kinds* of data can constrain it, if the model and data make it identifiable. That is the subject of Section 4.

Exercises for this lecture

Exercises 4, 6, numbered as in Appendix A of the full notes. Worked solutions are in Appendix B; attempt each one before reading its solution.

4. **[P] When the expectation does not exist.** A TOY main-sequence lifetime scales as $t \propto M^{-2.5}$. Let $X = M/M_\odot \sim \mathcal{N}(1, 0.1^2)$. (a) Estimate $\mathbb{E}[t]$ and $\mathrm{sd}(t)$ by the first-order delta method. (b) Add the second-order correction to the mean. (c) Is the exact $\mathbb{E}[X^{-2.5}]$ finite? What would a Monte Carlo estimate with 10^6 draws report, and why is that misleading? (d) Repeat (b) for a lognormal input, $\log X \sim \mathcal{N}(0, 0.1^2)$, and compare with the exact value.

6. **[P] How many stars until the floor?** With a shared offset $\sigma_\theta = 0.1$ and per-star scatter $\sigma_\epsilon = 0.2$, how many stars bring $\mathrm{sd}(\bar{Y})$ within 10% of its $N \to \infty$ limit? What, besides more stars, could reduce the limit?